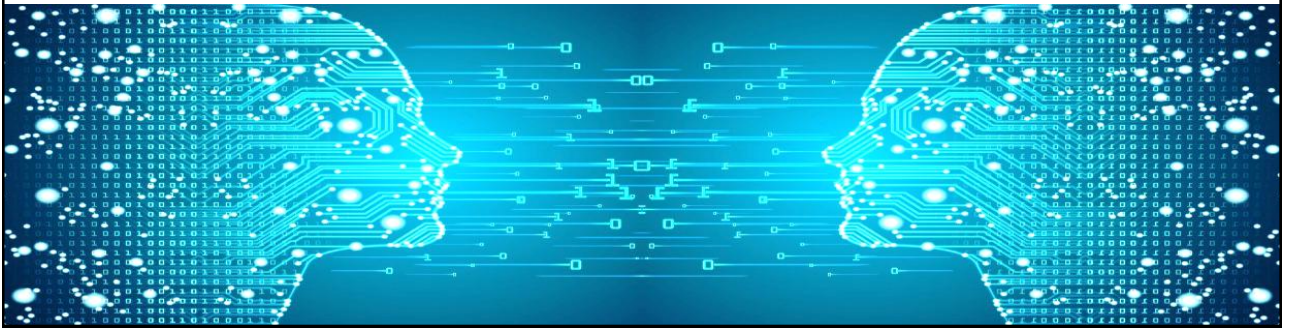
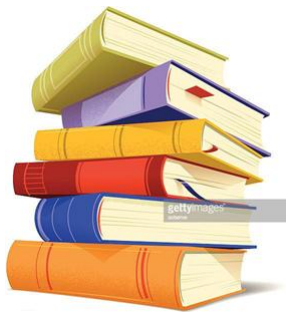


Machine Learning



Principal Component Analysis (PCA)



- Introduction
- Variance, covariance, eigenvalues, and eigenvectors
- PCA Workflow
- Applications of PCA

Why Do We Need PCA?

- High-dimensional data is difficult to visualize and analyze.
- Many features are strongly correlated, leading to redundant information.
- PCA:
 - Extracts the most informative directions in the data.
 - Reduces dimensionality while preserving most of the variability.
 - Helps denoise data and improve efficiency in machine learning.

Variance

- Variance measures how far the values of a variable spread around their mean. Larger variance means greater dispersion.

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$$

- Example:
 - Given data: $X = [2, 4, 6, 8]$
 - Number of values: $n = 4$
 - Mean: $\mu = \frac{2+4+6+8}{4} = 5$
 - Variance: $\sigma^2 = \frac{(2-5)^2 + (4-5)^2 + (6-5)^2 + (8-5)^2}{4} = \frac{9+1+1+9}{4} = 5$

Covariance

- Covariance measures how two variables change together.
- Positive covariance means they increase together.
- Negative covariance means one increases while the other decreases.
- Zero covariance means no linear relationship between the variables.

Covariance

- The population covariance between X and Y is computed as:

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})$$

- Example:

- Given two variables: $X = [2, 4, 6]$, $Y = [1, 3, 5]$
- Number of samples $n = 3$, $\bar{X} = 4$, $\bar{Y} = 3$
- $\text{Cov}(X, Y) = \frac{1}{3}((2 - 4)(1 - 3) + (4 - 4)(3 - 3) + (6 - 4)(5 - 3)) = \frac{8}{3} \approx 2.67$

Covariance Matrix

$$\Sigma = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_d) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \cdots & \text{Cov}(X_2, X_d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_d, X_1) & \text{Cov}(X_d, X_2) & \cdots & \text{Cov}(X_d, X_d) \end{bmatrix}$$

- A square matrix that summarizes the covariance between all pairs of variables.
- Diagonal elements represent variances of individual variables.
- Off-diagonal elements represent pairwise covariances.
- Provides a full view of the spread and correlation structure.
- PCA relies directly on this matrix.

Eigenvalues and Eigenvectors

- Eigenvectors represent directions of maximum variance in the data.
- Eigenvalues represent the amount of variance along each direction.
- Eigenvectors are orthonormal and form a new coordinate system.
- Sorting eigenvalues in descending order gives the principal components.
- The first principal component (PC1) captures the largest variance.

Orthonormal Vectors

- Vectors that are perpendicular to each other.
- Each has a magnitude (length) equal to 1.
- Necessary for forming a clean, non-overlapping coordinate system.
- PCA ensures the principal components are always orthonormal.
- This simplifies projection and distance calculations.

PCA Intuition

- PCA finds the axes that capture the most meaningful variation in the data.
- PCA rotates the coordinate system to align with these directions.
- The new axes are ordered by how much variance they capture.
- Reducing dimensionality means keeping only the top k axes.
- This preserves important information while removing redundancy.

PCA Workflow

1. Standardize the data.
2. Compute the covariance matrix.
3. Compute eigenvalues and eigenvectors.
4. Sort eigenvalues in descending order.
5. Select the top k principal components.
6. Transform the data into the new feature space.

Data Standardization

- Standardization sets mean to 0 and variance to 1 for each variable.
- Prevents variables with large magnitudes from dominating the PCA.
- Ensures all variables contribute fairly.
- Standardization is highly recommended unless all features share the same scale.
- PCA on unstandardized data often leads to misleading results.

Data Standardization

- The **standardization (z-score normalization)** formula is:

$$z_i = \frac{x_i - \mu}{\sigma}$$

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i$$

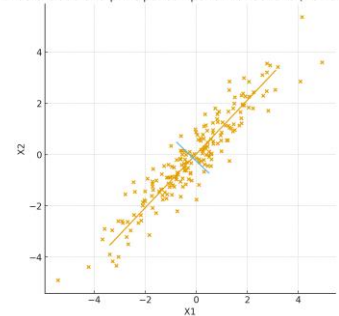
$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2}$$

- Meaning:
 - x_i is the original value of the variable.
 - μ is the mean of the variable.
 - σ is the standard deviation of the variable.
 - z_i is the standardized value.
- After standardization: $\text{Mean}[z] = 0$, $\text{Var}(z) = 1$

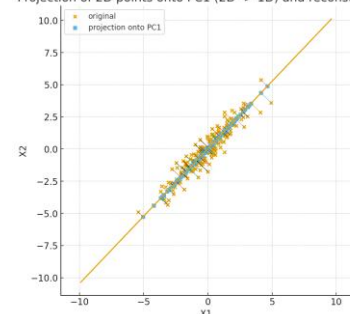
2D Example

- Consider 2D data points forming an elongated cloud.
- The longest direction of the cloud is PC1.
- The second longest orthogonal direction is PC2.
- Projecting the data onto PC1 compresses 2D into 1D while keeping most information.
- PCA essentially finds the axes that best fit the shape of the data.

2D data cloud and principal component directions (PC1 and PC2)



Projection of 2D points onto PC1 (2D -> 1D) and reconstruction



Numerical Example

- Suppose we have two features X and Y.
- Their covariance matrix indicates strong correlation.
- Eigenvalues: $\lambda_1 = 4.7$, $\lambda_2 = 0.3$.
- PC1 explains 4.7 out of 5 total variance, or 94 percent.
- Therefore, keeping only PC1 is enough for dimensionality reduction.

Explained Variance Ratio

- Explained Variance Ratio shows **the proportion of total variance explained by each PC**.

$$\text{Explained Variance Ratio}_i = \frac{\text{Variance of PC}_i}{\text{Total Variance}}$$

- Example: If PC1 has a ratio of 0.6, it means **60% of the total variance is captured by PC1**.
- High explained variance indicates that the PC captures a significant portion of the data's variability.

Cumulative Explained Variance

- Cumulative explained variance is the **sum of explained variance ratios for the first k PCs**:

$$\text{Cumulative explained Variance}_k = \sum_{i=1}^k \text{Explained Variance Ratio}_i$$

- Helps determine **how many PCs (k) to retain** for dimensionality reduction.
- Typical rule: retain enough PCs to explain **90–95% of the total variance**.

Interpreting PCA Loadings

- Loadings show how each original variable contributes to each PC.
- Large positive or negative loading values indicate strong influence.
- PC1 often indicates general trends or dominant behaviors.
- PC2 usually captures variations not explained by PC1.
- Loadings help explain the meaning of the new axes.

Applications of PCA

- Dimensionality reduction for high-dimensional data.
- Data visualization by reducing to 2D or 3D.
- Noise reduction by removing low-variance components.
- Preprocessing for machine learning to speed up training.
- Pattern discovery and feature extraction.

Limitations of PCA

- PCA captures only linear relationships.
- Sensitive to outliers.
- Not ideal when data is non-Gaussian or nonlinear.
- Hard to interpret when dataset has many variables.
- Requires scaling for meaningful results.

PCA Extensions

- Kernel PCA handles nonlinear structures.
- Sparse PCA improves interpretability by forcing many loadings to zero.
- Incremental PCA processes large datasets in batches.
- Robust PCA handles outliers.
- Randomized PCA is faster for very large matrices.

Summary

- PCA transforms data into a new coordinate system maximizing variance.
- Principal components are eigenvectors of the covariance matrix.
- Eigenvalues indicate how much variance each component captures.
- Dimensionality reduction is achieved by keeping only the top components.
- PCA is widely used for analysis, visualization, and preprocessing.

